

Lucky Verma

MACHINE LEARNING ENGINEER | AGENTIC SYSTEMS | KNOWLEDGE GRAPHS | RAG

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SUMMARY

Machine Learning Engineer with 4+ years building production agentic AI systems, knowledge graphs, and deep learning pipelines for large-scale data infrastructure. Expert in multi-agent orchestration (LangGraph, DSPy), RAG pipelines, entity resolution, model evaluation, and real-time inference serving on GPU clusters.

TECHNICAL SKILLS

AI / LLM	RAG, Agentic AI, LangGraph , DSPy , LangChain, vLLM, Fine-tuning (LoRA, RLHF/DPO), Tool Use, HF Transformers
ML / DL	PyTorch , TensorFlow, FSDP, DeepSpeed, SLURM Multi-GPU, Triton Kernels, Deep Learning , Computer Vision , NLP, CUDA
Infra	Knowledge Graphs (Apache AGE, FalkorDB), PostgreSQL, LanceDB, FastAPI , Docker , K8s, Terraform, AWS , Redis, Model Monitoring
Languages	Python , SQL, C++, JavaScript, Bash

WORK EXPERIENCE

Eccalon LLC (AI and federal data systems) MACHINE LEARNING ENGINEER	Nashville, TN Jun 2022 – Present
- Architected and deployed production deep learning + agentic AI workflows for regulated document analysis, evidence retrieval, and corporate ownership reasoning using LangGraph orchestration, DSPy optimization, and multi-step reasoning across federal and corporate data sources. - Engineered knowledge graph infrastructure (PostgreSQL + Apache AGE) with entity resolution across SAM.gov, SEC EDGAR, and GLEIF corporate registries, supporting multi-hop ownership-chain tracing with low-latency query response. - Built production document intelligence pipeline with GPU-accelerated embeddings (Ollama + NIM fallback), LanceDB vector search, and FalkorDB knowledge graph — reducing multi-source research synthesis from days of manual review to minutes across federal databases and regulatory documents. - Built model evaluation framework for LLM baselines, feature engineering, contamination checks, and multi-annotator validation in ownership and compliance-oriented reasoning workflows. - Deployed production RAG pipelines on AWS with multi-path reasoning, hybrid vector/keyword search, and source-grounded retrieval; built GPU-accelerated CV pipelines with PyTorch and CUDA. - Owned end-to-end ML infrastructure — Docker multi-service orchestration, Terraform IaC, CI/CD, model serving, monitoring, and multi-backend embedding fallback for production ML services.	
University of Maryland, Baltimore County GRADUATE RESEARCH ASSISTANT	Baltimore, MD Jan 2022 – May 2023
Developed ensemble 1D U-Net for biomedical signal classification (PPG → sleep staging), achieving 92% accuracy; built web-based data science learning platform using CV/ML pipelines with pandas and Streamlit.	
Vast Dream Group AI SPECIALIST	Mumbai, India (Remote) Jan 2021 – Aug 2021
Built recommendation engine using BART zero-shot classification (96% accuracy on MNLI); architected AI measurement systems for optics industry, contributing to \$1M+ revenue pipeline across 3 product lines.	
TandM Techlabs SOFTWARE DEVELOPER	Secunderabad, India Jun 2018 – Dec 2020
Built urban mobility platform with geospatial ML: Pandana accessibility metrics, Valhalla/OSMnx route optimization on PostGIS, cached with Redis and containerized on Kubernetes — serving real-time discovery for B2B logistics.	

KEY PROJECTS & OPEN SOURCE

Ownership Risk Evaluation Workbench Multi-source entity resolution LLM baseline evaluation Evidence tracing	
Built an evaluation workbench for corporate ownership reasoning, evidence tracing, and baseline comparison under regulated review workflows. Public release language is intentionally scoped until validation artifacts are ready.	
InfiniteContext-1B: Million-Token LLM Architecture DeepSeek-V3 MLA Distributed FSDP Consumer GPU inference	GitHub
Designing MLA-based LLM architecture targeting 93%+ KV cache reduction for million-token inference on consumer GPUs; implementing custom Triton kernels and FSDP distributed training pipeline on SLURM clusters. <i>[In progress]</i>	

EDUCATION

University of Maryland, Baltimore County Master of Science in Computer Science – GPA: 3.97	Baltimore, MD May 2023
DARPA Catalyst Award (\$50K) – Caselet, selected from 250+ teams for AI-powered national security workforce tools	
SRM University Bachelor of Technology in Electrical & Electronics Engineering – GPA: 3.9	Chennai, India May 2019